

Supplementary Material

A Computable Map of Treatment-Sequencing Evidence in HR+/HER2- Metastatic Breast Cancer

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Supplementary Table S1. Reporting Transparency Checklist

This study reports a graph-theoretic computational analysis of clinical trials and guidelines. No standard reporting guideline (TRIPOD, STROBE, PRISMA, CONSORT, STARD) is a perfect fit. We adopt a custom transparency checklist that combines PRISMA-2020 reporting items (relevant to evidence synthesis) with a graph-encoding addendum specific to the present method.

Item	Reported in	Notes
1. Title identifies study as evidence-synthesis	Title	“treatment-sequencing evidence”
2. Structured abstract	Abstract	300-word JCO CCI structure
3. Rationale	Introduction	Composition across non-overlapping trials
4. Objectives	Introduction	EFDPR + ODI
5. Eligibility criteria for trials	Methods 2.1, 2.2	16-trial pilot drawn from ESMO/ASCO reference lists
6. Information sources	Methods 2.1	ClinicalTrials.gov v2 + ESMO 2024
7. Search strategy	Methods 2.1	Per-NCT fetch list (Supplementary Table S2)
8. Selection process	Methods 2.2	Hand-curated against canonical primary publications
9. Data extraction process	Methods 2.2	Schema v1.0 (frozen)
10. Data items	Methods 2.2	Schema fields per Table S2
11. Risk-of-bias assessment	Discussion	Reviewer-cycle-disclosed corpus-selection bias
12. Effect measures	Methods 2.5	EFDPR; ODI
13. Synthesis methods	Methods 2.5	Concordance algorithm
14. Reporting bias assessment	Limitations	Pilot-scale only
15. Certainty assessment	Discussion	Pre-registered tolerance levels
16. Study selection results	Results 3.1	16/16 trials retrieved
17. Study characteristics	Table S2	Hand-curated schema fills
18. Risk of bias	Limitations	Disclosed
19. Results of individual studies	DAG edges	Trial primary endpoint per edge
20. Results of syntheses	Results 3.2–3.5	EFDPR, ODI, sensitivity
21. Reporting biases	Discussion	Reviewer-cycle history
22. Certainty of evidence	Discussion	Pre-registered exact test failed to reject
23. Discussion	Discussion	Three caveats + three positives + outlook

Item	Reported in		Notes
24. Limitations	Discussion		Pilot scale, hand-curation, single guideline
25. Implications	Discussion		Outlook
26. Registration and protocol	Methods 2.6, pre-reg		<code>docs/prereg.md</code>
27. Support	Funding	state-ment	No specific funding
28. Competing interests	Conflicts	state-ment	None declared
29. Availability of data, code, materials	Statements		MIT licence, GitHub URL, Zenodo DOI at release
G1. Graph node space	Methods 2.3		(prior-state, biomarker) tuples
G2. Graph edge construction	Methods 2.3		Trial $\rightarrow$ edge from required-state node to post-trial-state node
G3. Guideline-tree encoding	Methods 2.4		16-node encoding in Supplementary Table S3
G4. Concordance algorithm	Methods 2.5		State-superset + biomarker tolerance + drug match + temporal precedence
G5. Tolerance levels	Methods 2.5		Strict / ESCAT / liberal, all pre-registered
G6. Reproducibility (seed + commit hash)	Methods 2.7		Seed 20260516; v1.0.0 commit

Supplementary Table S2. Structured Trial Extractions

The hand-curated schema v1.0 mapping for each of the 16 pivotal trials is released as the machine-readable file `data/processed/trials_structured.json` in the repository. Each record contains the fields listed in Methods 2.2, the source primary publication, and the canonical ClinicalTrials.gov NCT identifier. The Round 1 internal review identified two NCT-misidentification errors in the original draft (NCT03997123 was incorrectly labelled as postMONARCH; NCT04032080 was incorrectly labelled as INAVO120); these were corrected to NCT05169567 (postMONARCH) and NCT04191499 (INAVO120), with TROPiCS-02 (NCT03901339) and OlympiAD (NCT02000622) added to the corpus.

Supplementary Table S3. ESMO HR+/HER2- mBC Decision-Tree Encoding

The hand-curated 16-node encoding of the ESMO 2024 update HR+/HER2- subset is released as `data/processed/esmo_decision_tree.json`. Each node records: node identifier, patient sub-state, biomarker profile, recommended drug classes with strength-of-recommendation grade, cited pivotal trials, and year of decision-tree introduction.

Supplementary Table S4. ODI Biomarker-Token Vocabulary

The operational-token vocabulary used to compute the operationalization discordance index is released as `analysis/06_compute_odi.py`. Each biomarker variable (HER2-low, ESR1mut, PIK3CAmut, AKT-pathway, prior-CDK4/6i) maps to a per-trial set of operational tokens drawn from the canonical primary publication.

Supplementary Text 1. Pre-registration deviations

The following pre-registration deviations are reported in the spirit of the pre-registration commitment in `docs/prereg.md`:

- **Corpus expansion from prototype (6 trials) to pilot (16 trials).** The pilot expansion was anticipated in the prereg and is not a deviation. Trial selection was based on the ESMO/ASCO mBC guideline reference lists; a systematic search of ClinicalTrials.gov for HR+/HER2- mBC pivotal trials is deferred to the production-scale extension.
- **Trial-identification corrections.** The original draft erroneously mapped NCT03997123 to postMONARCH and NCT04032080 to INAVO120; the corrections are NCT05169567 and NCT04191499, both verified against ClinicalTrials.gov briefTitle and condition fields. TROPiCS-02 and OlympiAD were added during the same correction pass to cover decision nodes G14 (post-CDK4/6i+post-chemo sacituzumab) and G16 (gBRCAm PARPi) that the prereg’s narrative implied but the prototype did not include.
- **Secondary outcomes S2, S3 reported descriptively.** The prereg commits to a Benjamini-Hochberg correction at $q = 0.05$ across S1, S2, S3. In the present pilot S1 (ODI) is reported as a per-biomarker mean Jaccard distance; S2 (temporal lag) and S3 (subgroup coverage) are reported descriptively because the small corpus precludes a meaningful formal test.
- **Primary CI: Clopper-Pearson alongside bootstrap.** The prereg specified a 1,000-iteration bootstrap CI. We additionally report the Clopper-Pearson exact CI, which is the recommended primary CI in this small- n regime and which is reported alongside the bootstrap CI in Table 2.
- **NCCN sensitivity analysis deferred.** The prereg flagged NCCN as a sensitivity-analysis source; this was deferred to the production-scale extension and is disclosed as a deviation.